

TIGP CLASS BOARD 2026 – laboratory instrument carrier (ESP32-S3 dev board) with NMR console, rev B (v0.7, 2026-09-13)

Design brief v0.6 + notes/2026-09-13-nmr-respec-proposal.md (v0.7) + hardware/docs/design-decisions.md. Interface nets between sheets are global labels; rails are power symbols; AGND meets GND only at NT1.

Stack-up: L1 signal/power, L2 GND, L3 GND, L4 signal/power. Board 180 x 100 mm, 4 x M3. Bottom side: three 2x20 straight male headers (J6 at x 12, J7 at x 90, J8 at x 168 mm) carry the front panel (180 x 100, 17 SMA) flat on the back of the board. Rear edge: USB-C, 5 V jack, relay contacts, isolated inputs, external po

A base_mcu

BASE: dev-board socket, buses, expansion, star point

File: sheets/base_mcu.kicad_sch

b2_power

B2 (C): power entry, ± 12 V, LDOs

File: sheets/b2_power.kicad_sch

b1_inputs

B1 (A): $8 \times +10$ V inputs, ADS8688

File: sheets/b1_inputs.kicad_sch

b3_outputs

B3 (B): DAC8563 + OPA2192 -> A01/A02

File: sheets/b3_outputs.kicad_sch

b4_switching

B4 (C): 4 relays, 2 isolated inputs

File: sheets/b4_switching.kicad_sch

b5_dio_trig

B5 (B): TCA9535 → 8 TTL DIO + relays, 2 fast outs, TRIG

File: sheets/b5_dio_trig.kicad_sch

front_panel_link

LINK: 3 x 2x20 to the front panel (bottom side)

File: sheets/front_panel_link.kicad_sch

c_switch

C: external power input, DRV8871 field-cycling H-bridge, polarizer MOSFET switch

File: sheets/c_switch.kicad_sch

nmr_rx

NMR RX (A): tank, LNA, blanking, I/Q mixer, IF → ADC ch 7/8

File: sheets/nmr_rx.kicad_sch

nmr_tx

NMR TX (B): AD9834 DDS, reconstruction filter, OPA564 power stage → TX

File: sheets/nmr_tx.kicad_sch

GPIO map v0.7 (interface contract, brief 4.2 + proposal 8.2)

GPIO	Net	Function	Section
12/11/13	SPI_SCLK/MOSI/MISO	SPI2 IO_MUX, 33 R at source (ADC, DAC, DDS)	base
10	CS_ADC	AD58688 /CS	A
5	CS_DAC	DAC8563 /SYNC (via 74HCT125)	B
1/2	I2C_SDA/SCL	400 kHz: OLED 0x3C, TCA9535 0x20, S15351A 0x60, Qwiic J4 + panel Qwiic	base
16/17	OPTO_IN1/2	6N137 outputs, 1 k pull-up, active LOW	C
18/21	FAST_OUT1/2	74HCT125 -> 49.9 R -> panel link -> 2 fast SMAs on the panel	B
38/39	TRIG_IO / TRIG_DIR	74LVC1T45: DIR 1 = output to the panel SMA	B
41/42	DDS_FSYNC / DDS_PSEL	AD9834 frame sync / phase-register select	B
40	TX_EN	OPA564 enable = transmit gate (10 k pull-down)	B
8	RX_BLANK	DG419 receiver blanking (pull-up: blanked)	A
9/14	HB_IN1/2	DRV8871 H-bridge inputs (10 k pull-downs)	C
47	FET_GATE	UCC27517 -> A0D4184A polarizer switch	C
TCA9535 P00-P07	DIO1..8	74AHC1541 -> 1 k -> panel link -> panel TTL strip	B
TCA9535 P10-P13	RLY_IN1..4	A03400A relay gates (10 k pull-down)	C
4/6/7/15	GPIO4/6/7/15	free -> 2x10 expansion header	base
19/20	USB_DN/DP	native USB from the carrier USB-C	base
43/44	GPIO43/44	header: LED_WIFI/ACT via JP1/JP2; TCXO option	base
3, 0/45/46, 35-37, 48	-	strapping / PSRAM / on-board LED: unused	-

Rails (brief 4.1, updated per design-decisions D-02/D-04; +VEXT 7-18 V external input on the C sheet)

Rail	Source	Budget	Feeds
+5V_RAW	USB-C or jack via polyfuse + TVS + LM66100	<= 1.5 A (est. 1.15–1.35 A)	dev board, relays (4 x 71 mA), 5 V logic, LDO, DC–DC
+3V3	AMS1117–3.3	<= 500 mA (est. < 150 mA)	logic side, OLED, optos, Qwiic
+12V / –12V	B0512S–2WR3 x2, AGND–referenced	166 mA each (est. 40 / 20 mA incl. bleeders)	78L05, OPA2192, clamps, OPT
+5VA	78L05G from +12V	<= 100 mA (est. 17 mA)	AD58688 / DAC8563 AVDD
AGND	star point NT1 (net tie) to GND	–	L1/L4 analog copper, panel copper (through J6)

Reference ranges: base 1–99, B1 100–199, B2 200–299, B3 300–399, B4 400–499, B5 500–599, NMR RX 700–799, NMR TX 800–899, coil switches / external power 900–999 (owner-zone DRC uses the Block field: A / B / C).

Student gapped copies (D6) are derived from these sheets: see [hardware/student/](#) and [docs/student-deletions.md](#).

ESP32-S3 laboratory instrument carrier + NMR console: base + 3 student sections (A input)		
TIGP class board 2026		
Sheet: /		
File: class-board.kicad_sch		
Title: TIGP class board 2026 – root		
Size: A3	Date: 2026-09-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 1/11

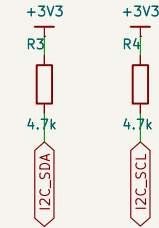
BASE – ESP32–S3 dev-board socket, I2C, expansion header, star point

Dev board: Jinhua #40729, ESP32–S3–DevKitC–1 v1.1 pin order (Espressif user guide). 3V3 pins NOT connected (dev board LDO must not fight the carrier's AMS1117). Strapping pins I00/I045/I046 left open; I03 is unused as well (v0.7: it was DI08 – the strapping question is closed by not using it). I035–37 (octal PSRAM) and I048 (on–board WS2812) unused. GPIO map v0.7 (re–spec 8.2, interface contract): I041 DDS_FSYNC, I042 DDS_PSEL, I040 TX_EN, I08 RX_BLANK, I09/I014 HB_IN1/HB_IN2, I047 FET_GATE, DI01–8 and RELAY1–4 moved to the TCA9535 I2C expander (B5), which frees I04/6/7/15 to the expansion header. I2C addresses: TCA9535 GPIO expander 0x20 (A0 = A1 = A2 = GND, B5), Si5351A clock generator 0x60, OLED 0x3C. 400 kHz, 4.7 k pull–ups here; Qwiic J4 inside the box and a second Qwiic on the front panel (through the link). Socket row spacing 25.4 mm ASSUMED from the 28 mm–wide clone (HANDOVER 2026–09–05) – MEASURE a #40729 before ordering (docs/design–decisions.md D–12). SPI2 on IO_MUX pins: 33 R series at the source on SCLK/MOSI (here) and on SDO at the ADC (B1). All inter–sheet signals are global labels; rails are power symbols.

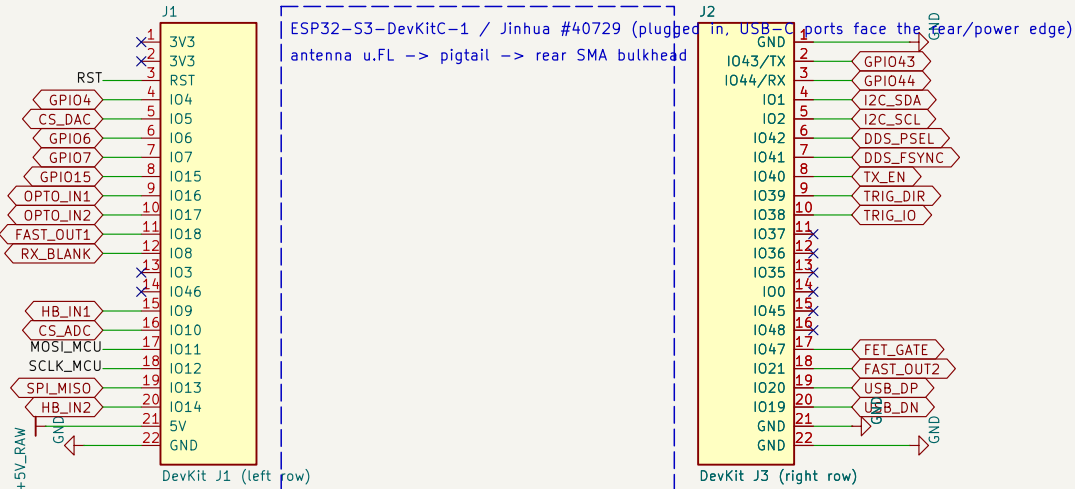
GPIO map v0.7 (re–spec 8.2) – interface contract

GPIO	Net	Function	Block
12 / 11 / 13	SPL_SCLK/MOSI/MISO	SPI2 on IO_MUX, 33 R at source	base
10 / 5	CS_ADC / CS_DAC	ADS8688 /CS; DAC8563 /SYNC	B1 / B3
1 / 2	I2C_SDA / I2C_SCL	400 kHz: 0x20, 0x60, 0x3C	base
16 / 17	OPTO_IN1 / 2	6N137 outputs, active LOW	B4
18 / 21	FAST_OUT1 / 2	74HCT125 -> 49.9 R -> panel link	B5
38 / 39	TRIG_IO / TRIG_DIR	74LVC1Y45 (1 = out to the SMA)	B5
41 / 42	DDS_FSYNC / DDS_PSEL	AD9834 sync / phase select	NMR TX
40 / 47	TX_EN / FET_GATE	OPA564 enable; polarizer FET	TX / B4
8	RX_BLANK	receiver blanking switch	NMR RX
9 / 14	HB_IN1 / HB_IN2	DRV8871 field–cycling bridge	B4
4 / 6 / 7 / 15	GPIO4/6/7/15	freed -> expansion header J5	base
43 / 44	GPIO43 / GPIO44	LED_WIFI/ACT via JP1/JP2; hdr	base
expander	DI01–8, RLY_IN1–4	P00–P07 -> 541: P10–P13 relays	B5 / B4
0/45/46, 3, 48	–	strapping / unused / PSRAM 35–37	–

I2C 400 kHz pull–ups (OLED 0x3C + Qwiic J4 inside + the panel Qwiic)

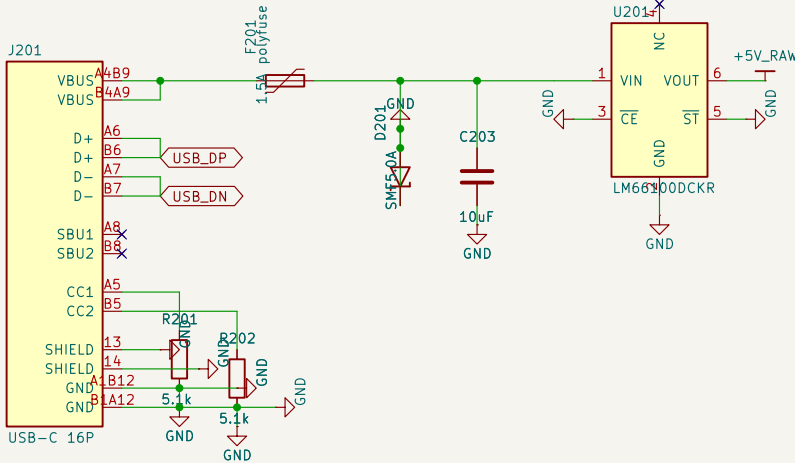


PWR_FLAGS: GND and AGND are references without a power_out driver

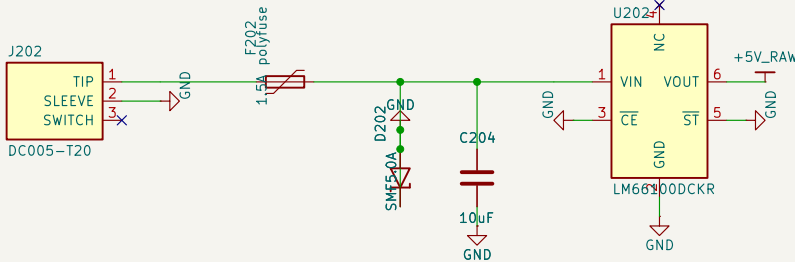


B2 – POWER: USB-C / 5 V jack entry, ideal-diode ORing, +–12 V isolated modules, +5VA and +3V3 LDOs

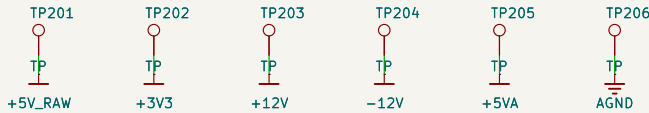
Entry: 1.5 A polyfuse -> SMF5.0A TVS -> LM66100 ideal diode per input (CE = GND, always on; the higher source wins, reverse current blocked). Drop ~0.18 V typ at 1.2 A vs 0.5 V for the Schottky ORing (D–02).
+–12 V: two unregulated B0512S–2WR3 (10 % minimum load rule!): +12 V rail loaded by 78L05 + bleeder, –12 V rail by OPA2192 + 3 x 2.2k bleeder (16 mA) so both stay inside the +–15 % envelope (datasheet p.3/4).
78L05G (SOT–89, 350 mW max): +12 V -> +5VA for ADS8688/DAC8563 AVDD; ~17 mA -> 0.12 W. AMS1117–3.3: 22 uF output required for stability (datasheet ‘Stability’); ceramic used, tantalum drop–in (3216) if the rail rings.
All power–module outputs are referenced to AGND (–12 V module: +Vo = AGND). GND and AGND meet only at NT1 (base sheet).



CC1/CC2 5.1k = UFP sink, 5 V. Shield pads to GND directly (no chassis).

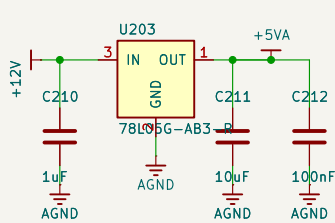
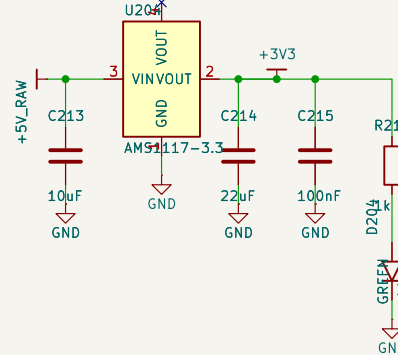
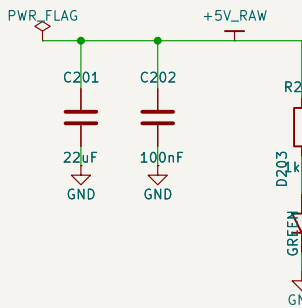


Jack: 5 V DC, centre +, 5.5/2.1 mm, 1.5–3 A. Switch pin unused.



Rail test points

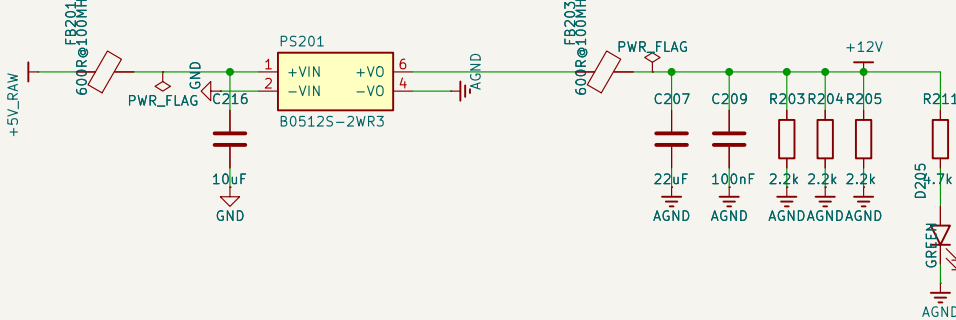
+5V_RAW rail (both LM66100 outputs): bulk 22 uF + 100 nF, green LED (2.9 mA); PWR_FLAG = ideal diodes drive the rail



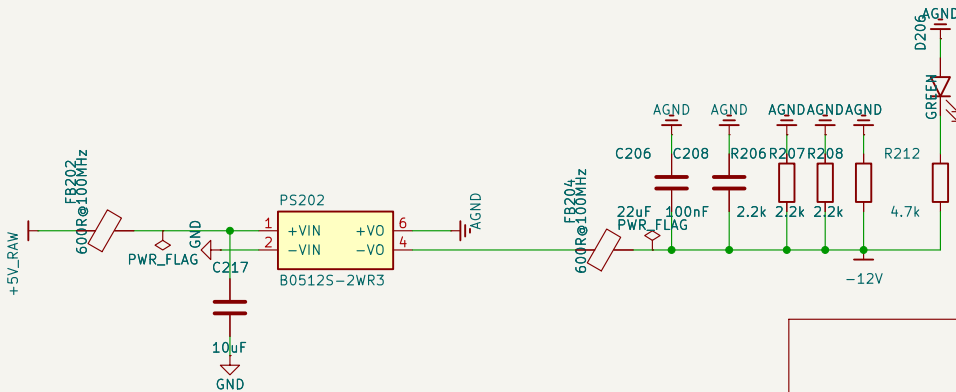
+5VA: 78L05G from +12V (dropout 1.7 V), ADC/DAC analog supply, AGND–referenced: 0.12 W at 17 mA

+3V3: AMS1117–3.3 (SOT–223), 22 uF ceramic out (tantalum 3216 drop–in if needed), pin 4 = tab (same VOUT net on the footprint)

Isolated 2 W modules: input 4.5–5.5 V, 454 mA full load, ripple 80 mVp–p max, Cout <= 560 uF, 1.5 kVDC isolation (YLPTec datasheet)



+12V: FB + 22 uF + 100 nF at the module: 3 x 2.2k = 16 mA bleeder guarantees the 10 % minimum load with 78L05 load included



–12V: module #2 with +Vo on AGND: FB in the –Vo line: 3 x 2.2k = 16 mA bleeder guarantees the 10 % minimum load with 78L05 load included

USB-C 5V jack + 5 V jack entry, ideal-diode ORing, +–12 V isolated modules, +5VA and +3V3 LDOs		
TIGP class board 2026		
Sheet: /b2_power/		
File: b2_power.kicad_sch		
Title: B2: power entry, ORing, isolated +–12 V, LDOs		
Size: A3	Date: 2026–09–07	Rev: A
KiCad E.D.A. 10.0.3		Id: 3/11

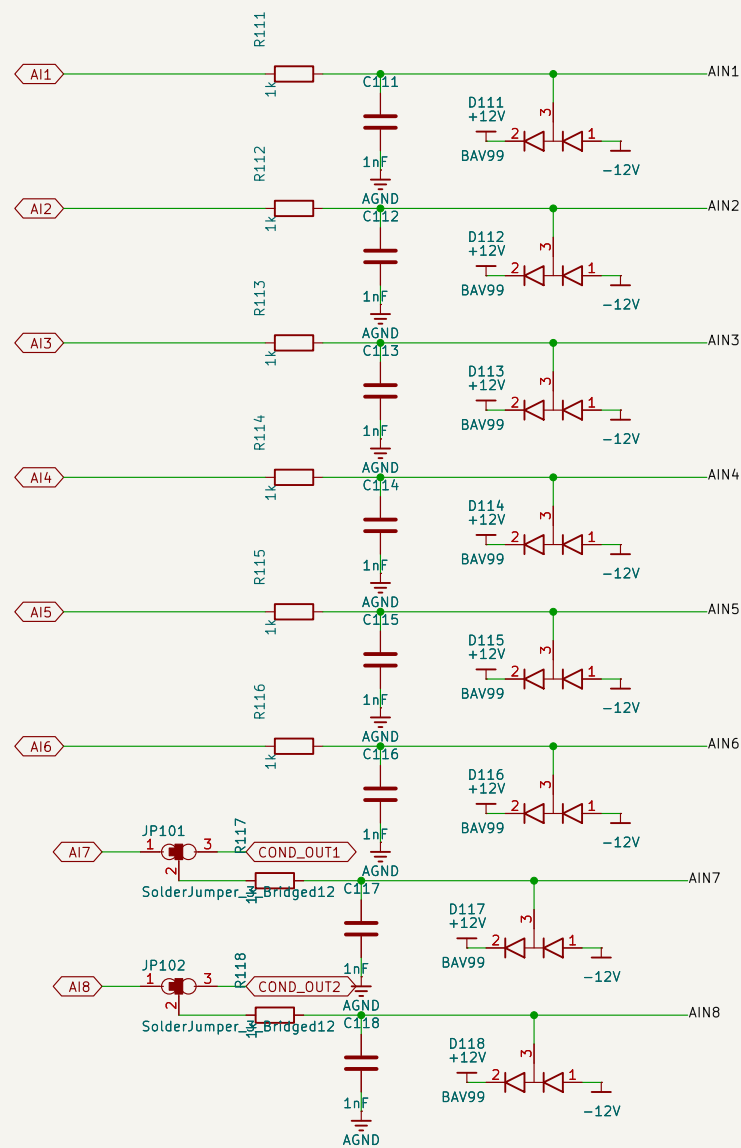
B1 – PRECISION INPUTS: 8 x (1 k / 1 nF / BAV99 clamp) -> ADS8688 16-bit SAR ADC

Input network per channel: 1 k series (limits clamp current to 36 mA at $+50$ V), 1 nF to AGND (f-3dB ~ 160 kHz), BAV99 clamps to the ± 12 V rails. ADC input $+10.24$ V range, 1 M Ω input impedance, abs max $+20$ V.

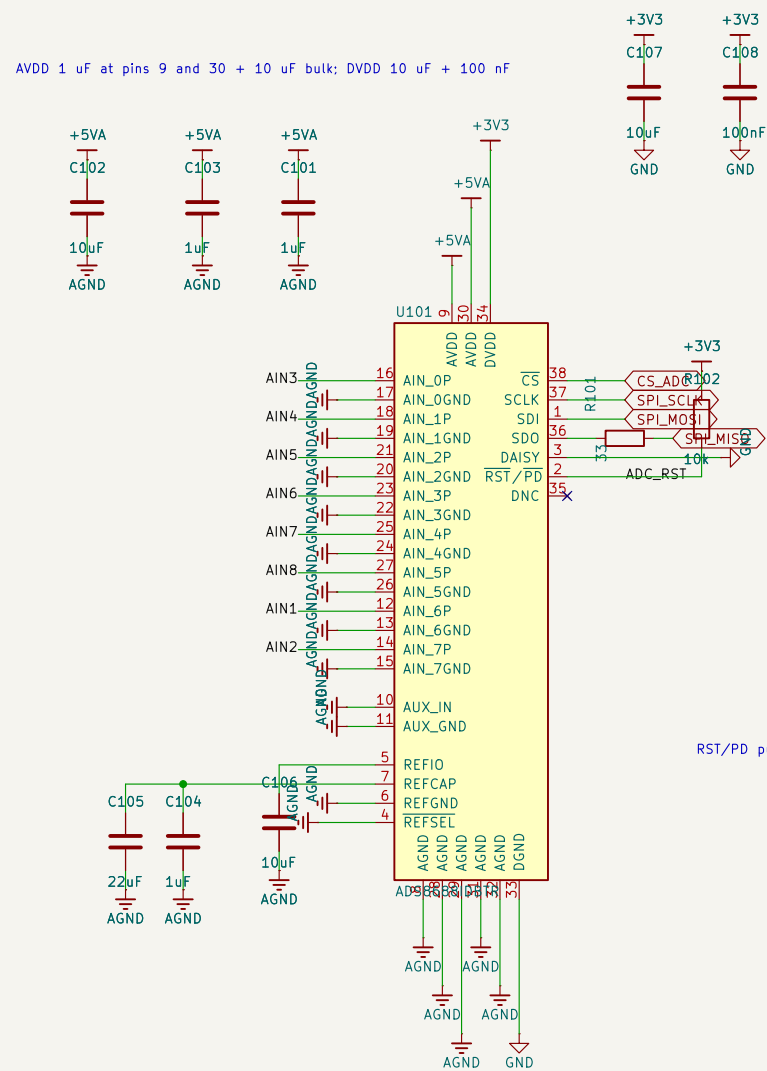
ADS8688 decoupling per datasheet 11.1: AVDD 1 uF at each pin + 10 uF; REFCAP 1 uF + 22 uF (min 10 uF) direct at pins; REFIO 10 uF (internal 4.096 V ref, REFSEL low); DVDD 10 uF + 100 nF. DAISY = GND, RST/PD pulled to DVDD.

Channel mapping (firmware table, D-19): AI1->AIN_6, AI2->AIN_7, AI3->AIN_0, AI4->AIN_1, AI5->AIN_2, AI6->AIN_3, AI7->AIN_4, AI8->AIN_5 (package order = planar fan-in from the link, no crossings). AI7/AI8 switch to the OPT chain with JP101/JP102 (default: SMA path).

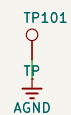
AGND = analog reference copper (L1/L4 pours), joined to GND only at NT1 (base). SDO has a 33 R source resistor.



8 identical input networks: 1 k $\rightarrow$ node (1 nF to AGND, BAV99 clamps to the rails) $\rightarrow$ AINn $\rightarrow$ ADC channel per the D-19 table



RST/PD pulled up to DVDD (10k); firmware may pull it low for reset/power-down



REFSEL low = internal 4.096 V reference; REFCAP 1 uF + 22 uF, REFIO 10 uF (ADS8688 datasheet 8.3.3 / 11.1)

AI1..AI8 -> 1k / 1nF / BAV99 clamp -> ADS8688 AIN_0..7; ch 7/8 solder jumpers to the OPT cha

TIGP class board 2026

Sheet: /b1_inputs/
File: b1_inputs.kicad_sch

Title: B1: eight ± 10 V inputs, ADS8688

Size: A3

Date: 2026-09-07

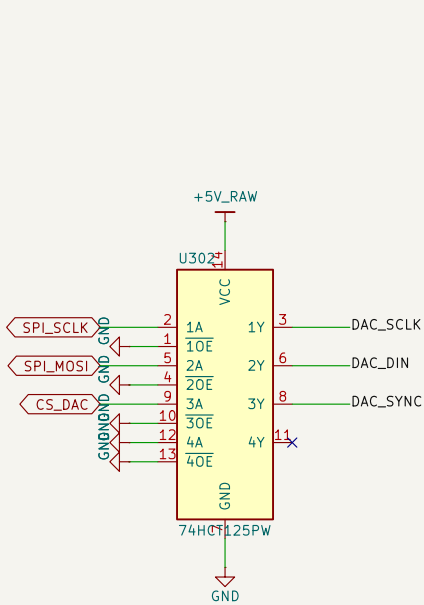
Rev: A

KiCad E.D.A. 10.0.3

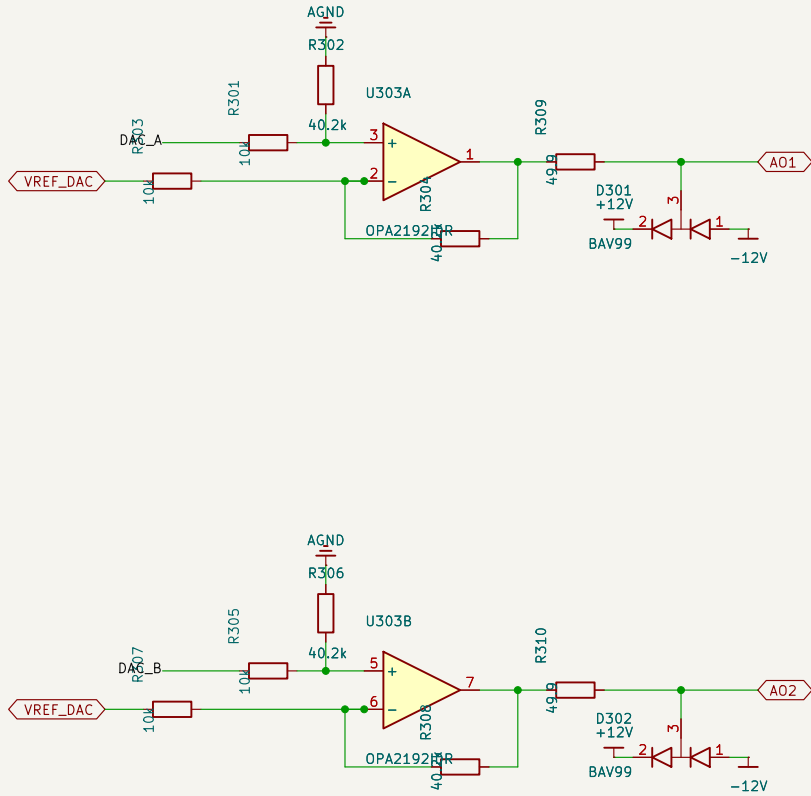
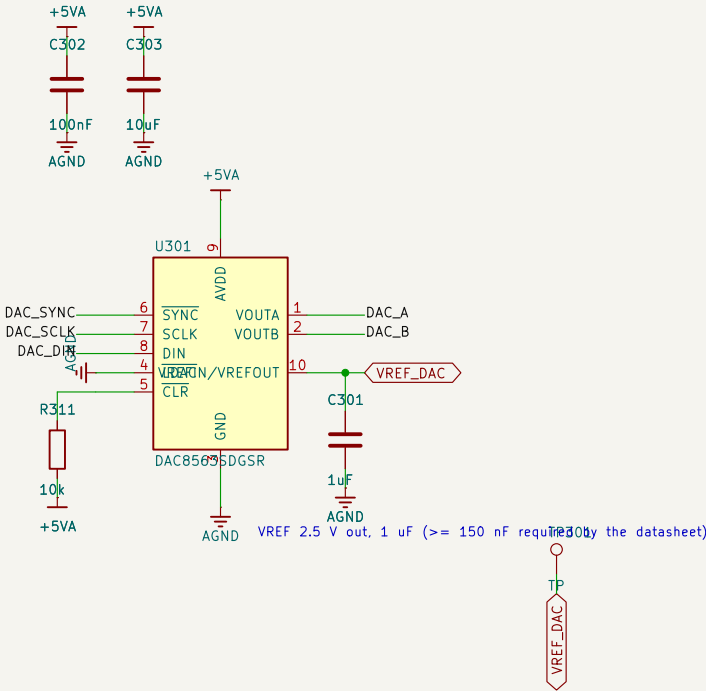
Id: 4/11

B3 – SIGNAL GENERATION: DAC8563 (0–5 V, internal 2.5 V ref) –> OPA2192 difference amplifiers –> A01/A02 = +-10 V

DAC8563 logic thresholds are 0.7 x AVDD = 3.5 V at AVDD = 5 V, so a 74HCT125 (TTL-compatible inputs, 5 V outputs, tPD ~20 ns equal on all three lines) translates SCLK/DIN/SYNC (D–05).
AO_n = 4 x (DAC_n – VREF): R1 = R3 = 10 k, R2 = R4 = 40.2 k (1 %, gain 4.02): 0..5 V in -> -10.05..+10.05 V out; offset/gain calibrated in firmware via AI loopback. VREF drives 2 x 10 k (0.5 mA of the +-20 mA capability).
Output: 49.9 R back-termination, BAV99 clamps to +-12 V. OPA2192 short-circuit current +-65 mA: into a 50 R termination the swing is limited to ~+-6 V; full +-10 V into >= 1 k loads.
Gain = 2 is the DAC default once the internal reference is enabled by firmware (reset default: reference off, gain 1). /LDAC = AGND (synchronous), /CLR pulled up to +5VA.



SPI -> 5 V level shift for the DAC (three channels, 4th input grounded)



$$AO_n = (1+R2/R1) \times R4/(R3+R4) \times DAC_n - (R2/R1) \times VREF = 4.02 \times DAC_n - 4.02 \times 2.5 \text{ V}$$

B4 – MAINS-CAPABLE SWITCHING: 4 x SPDT relay (10 A @ 277 V AC) on A03400A MOSFETs; 2 x 6N137 isolated inputs, 5–24 V

Relay drive: A03400A (48 mOhm) instead of a Darlington: the 5 V coil gets the full +5V_RAW (JQC-3FF/005-1ZS pull-in <= 3.75 V, coil 70 ohm = 71 mA per relay, 4 x 71 = 285 mA from +5V_RAW). 1 k gate series, 1N4148W flyback across the coil. Yellow LED shows coil on.

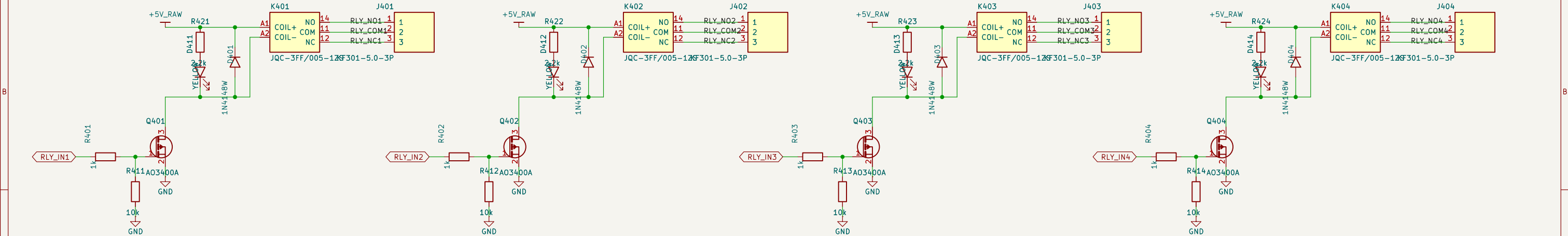
v0.7: the gates are driven by RLY_IN1..4 from the TCA9535 I2C expander (B5, address 0x20), not by a GPIO. The 10 k gate pull-down is what keeps every relay OFF while the ESP32 boots and while the expander powers up with all its ports as inputs (high-impedance) – it must not be omitted.

MAINS-capable channel: relay 10 A @ 277 V AC; board rating 250 V AC 5 A MAX per channel, load fused <= 5 A; MAINS net class: 5 mm to all other copper on every layer, F.Cu only, no vias, >= 3 mm tracks; terminal directly behind its relay on the rear edge; coil 70 ohm = 71 mA per relay (4 x 71 = 285 mA from +5V_RAW).

Contacts: COM/NO/NC to the 3P terminal, NO = pad 1, COM = pad 2, NC = pad 3. Relay pins are the footprint pad names (EN 50005): A1/A2 = coil, 11 = COM, 12 = NC, 14 = NO. The coil-to-contact inside the relay is the manufacturer's isolation (1.5 kV test) and is excluded from the 5 mm rule; mains is switched only with the contacts.

Isolated input: series 1N4148W (reverse blocking) -> 220 R -> Q1 pass transistor with Q2 sensing 0.65 V across Rs (100 R || 1 k = 91 R); LED current ~6.5–7 mA from 5 V to 24 V (6N137 needs 5–15 mA). At 24 V Q1 dissipates ~125 mW (MMBT5551 300 mW).

6N137: VCC/VE = +3V3, 1 k pull-up; output LOW when the input is driven (inverting). The isolated side nets (ISO1_*, ISO2_*) have a 2.5 mm clearance class and no copper of any other net underneath.



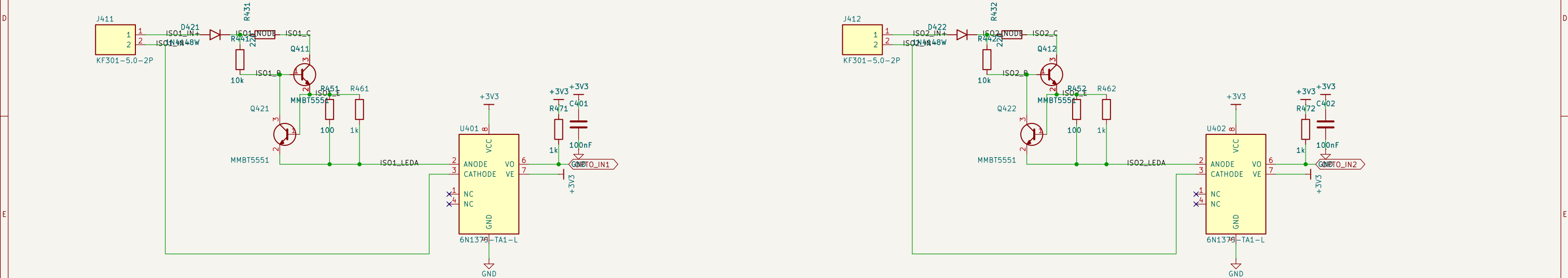
J401: 1 = NO, 2 = COM, 3 = NC MAINS: 250 V AC 5 A MAX, load fused <= 5 A J402: 1 = NO, 2 = COM, 3 = NC MAINS: 250 V AC 5 A MAX, load fused <= 5 A J403: 1 = NO, 2 = COM, 3 = NC MAINS: 250 V AC 5 A MAX, load fused <= 5 A J404: 1 = NO, 2 = COM, 3 = NC MAINS: 250 V AC 5 A MAX, load fused <= 5 A

K401: SPDT 10 A @ 277 V AC / 28 V DC, coil 5 V 70 ohm (71 mA); terminal directly behind its relay on the rear edge; coil 70 ohm = 71 mA per relay (4 x 71 = 285 mA from +5V_RAW). K402: SPDT 10 A @ 277 V AC / 28 V DC, coil 5 V 70 ohm (71 mA); terminal directly behind its relay on the rear edge; coil 70 ohm = 71 mA per relay (4 x 71 = 285 mA from +5V_RAW). K403: SPDT 10 A @ 277 V AC / 28 V DC, coil 5 V 70 ohm (71 mA); terminal directly behind its relay on the rear edge; coil 70 ohm = 71 mA per relay (4 x 71 = 285 mA from +5V_RAW). K404: SPDT 10 A @ 277 V AC / 28 V DC, coil 5 V 70 ohm (71 mA); terminal directly behind its relay on the rear edge; coil 70 ohm = 71 mA per relay (4 x 71 = 285 mA from +5V_RAW).

v0.7 gate drive: RLY_IN1..4 = TCA9535 P10..P13 (expander U505 on B5, I2C address 0x20) through the 1 k series resistor; the 10 k pull-down holds every relay off at boot and while the expander ports are still inputs.

MAINS design rules (class=board.kicad_dru): MAINS copper keeps 5 mm to every other net on every layer and 5 mm channel-to-channel, 2 mm between the nets of one channel, 2 mm to the board edge; F.Cu only, no vias, tracks >= 3 mm.

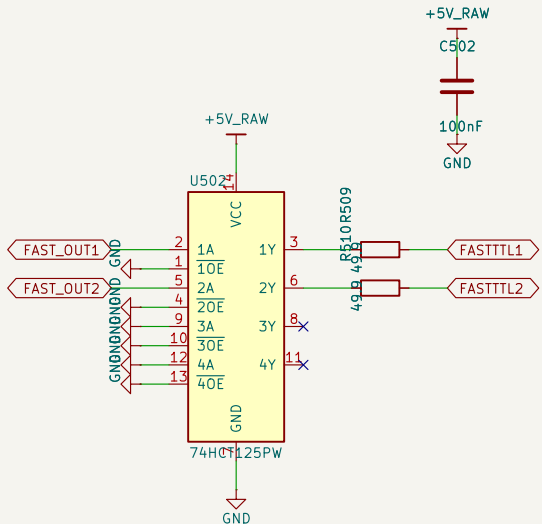
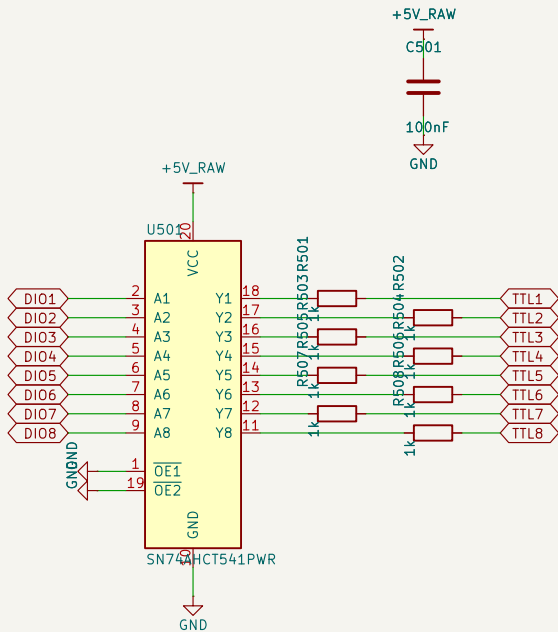
Coil current: 4 x 71 mA = 285 mA from +5V_RAW (the 125 ohm HK4100F coils drew 160 mA) – see the rails table on the root sheet.



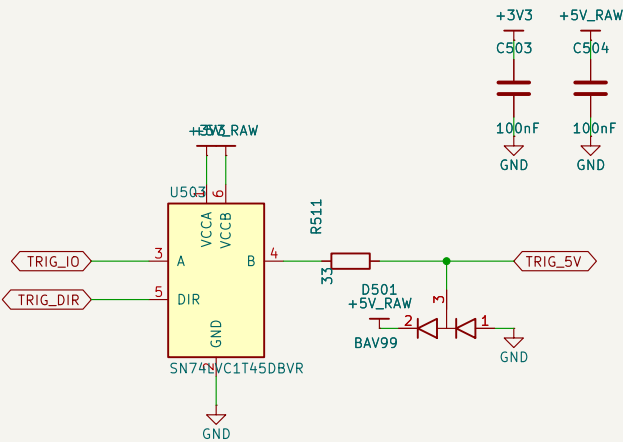
J411: 1 = IN+, 2 = IN- (5–24 V DC, isolated, 1.5 kV opto) J412: 1 = IN+, 2 = IN- (5–24 V DC, isolated, 1.5 kV opto)

B5 – DIGITAL I/O AND TIMING: 8 x 5 V TTL outputs, 2 fast outputs, bidirectional TRIG (panel SMA), 10 MHz TCXO option

74AHCT541 (TTL-compatible inputs accept 3.3 V) buffers DIO1..8 to 5 V; 1 k series per output limits a shorted output to 5 mA (students!). Outputs are static/slow (kHz): 1 k x 100 pF cable = 100 ns. v0.8: TTL1..8 and FASTTTL1/2 leave on the panel link, not on board-edge terminals.
v0.7: DIO1..8 keep their names but are no longer MCU pins — they come from the TCA9535 I2C expander U505 (address 0x20, A0 = A1 = A2 = GND), P00..P07. The 541 runs at +5V_RAW while the expander drives 3.3 V: the AHCT TTL thresholds (VIH 2.0 V) accept that. P10..P13 = RLY_IN1..4 (relay gates, B4), P14..P17 = spare on test point.
74HCT125: FAST_OUT1/2 (MCPWM/RMT) -> 49.9 R series -> the panel link -> the two fast SMAs on the panel; drive +-6 mA rated, so a 50 R termination gives ~1 V; fast outs are for high-impedance TTL loads.
TRIG: 74LVC1T45 A = TRIG_IO (3.3 V), B = 5 V side -> 33 R -> TRIG_5V -> panel SMA; DIR (GPIO39) 1 = output. Into a 50 R termination ~2.4 V typ (beyond the rated 32 mA), 5 V into high-Z. BAV99 clamps B to GND/+5V_RAW.
TCXO option (all DNP): clipped-sine 10 MHz -> 1 nF -> Schmitt buffer biased at VCC/2 -> 74LVC1G17 -> JP501 -> GPIO44 (open JP2 on the base sheet when used). VCONT at 1.65 V from a divider.



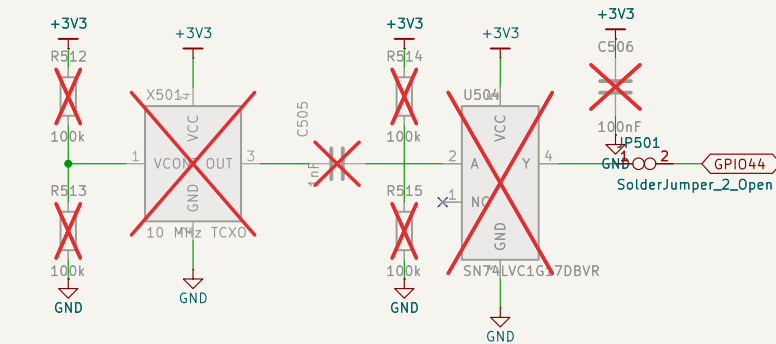
FASTTTL1/2 leave this sheet as global labels -> panel link J7 -> the two fast SMAs on the front panel (49.9 R series). The edge terminals J506/J507 are gone (v0.8).



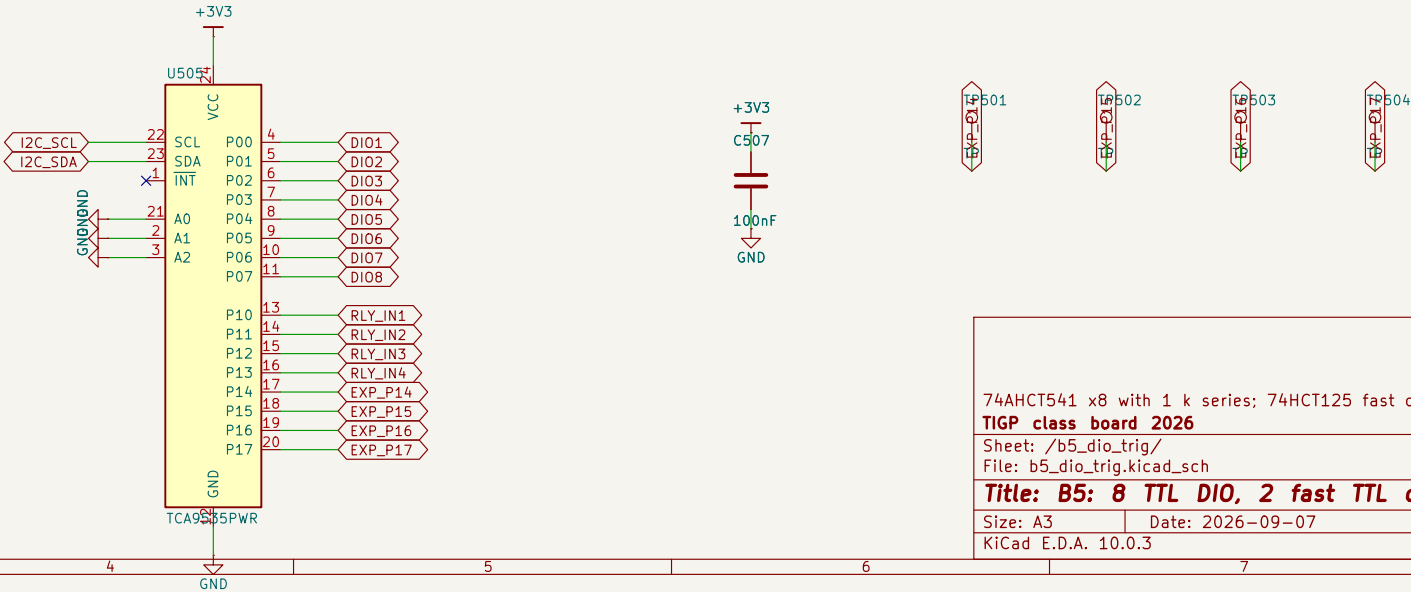
TRIG_DIR = 1: TRIG_IO drives the SMA (output); 0: SMA -> TRIG_IO (input, 5 V tolerant)

TTL1..TTL8 leave this sheet as global labels -> panel link J7 -> the 10-way screw-terminal strip on the front panel (TTL1..8 + 2 x GND). The board-edge terminals J501-J505 are gone (v0.8).

U505 TCA9535 (TSSOP-24, address 0x20, A0 = A1 = A2 = GND): the I2C port expander that replaced 12 direct GPIO in v0.7 — P00..P07 = DIO1..8 into the 74AHCT541 above, P10..P13 = RLY_IN1..4 to the relay gates (B4), P14..P17 spare on TP501..TP504. Every port is an input after power-up and after a reset: the relay gate pull-downs (B4) are what set the safe state.



10 MHz timebase option — all parts DNP by default (zero cost); JP501 open until a firmware use exists



74AHCT541 x8 with 1 k series; 74HCT125 fast outs with 49.9 R; 74LVC1T45 TRIG with 33 R + BAV99		
TIGP class board 2026		
Sheet: /b5_dio_trig/ File: b5_dio_trig.kicad_sch		
Title: B5: 8 TTL DIO, 2 fast TTL outs, bidirectional TRIG, TCXO option		
Size: A3	Date: 2026-09-07	Rev: A
KiCad E.D.A. 10.0.3		Id: 7/11

FRONT-PANEL LINK – three 2x20 straight male headers on the BOTTOM side of the main board (v0.8)

The panel stacks flat on the BACK of the main board (Decision #45). J6/J7/J8 are mounted on the BOTTOM side of the main board and hand-soldered from the top; they mate the panel's three female headers. Positions along the 180 mm edge: x ~ 12 (J6), x ~ 90 (J7), x ~ 168 (J8) mm.

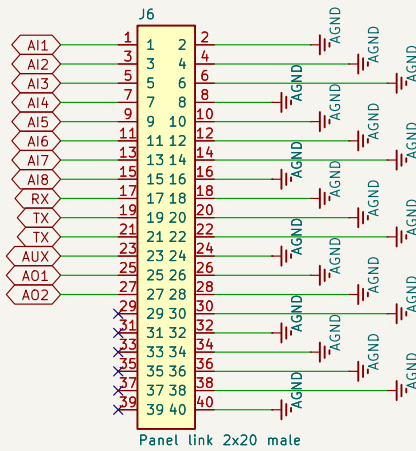
Odd pins = row 1 (signal), even pins = row 2 (return): every signal pin has its return, AGND or GND, on the adjacent even pin. J6 carries the analog signals (nearest the panel SMAs), J7 the digital ones, J8 the rails and the spare GPIO.

LED_PWR is +3V3 (always on; the LED series resistor sits on the panel). LED_WIFI / LED_ACT come from GPIO43 / GPIO44 through JP1 / JP2 (base sheet). GPIO4/6/7/15/43/44 on J8 are the same nets as on the expansion header J5. TX is on two pins (19 and 21) so the coil current has two paths.

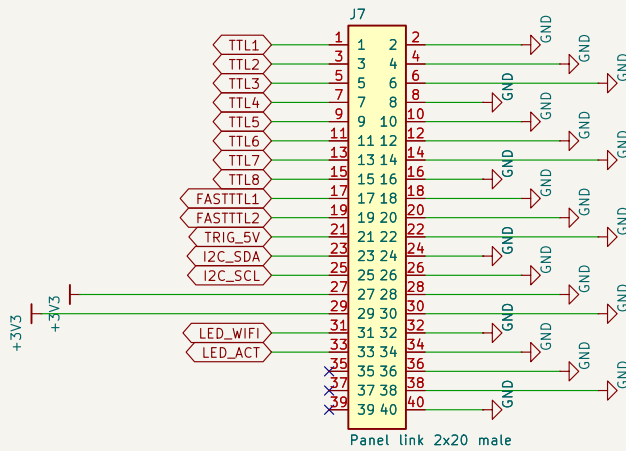
Pins marked spare are not connected on the main board; on the panel they reach labelled test pads, so a later panel part needs no new main-board revision. Panel copper is AGND and meets GND only at the star point NT1 on the main board.

The three headers (and the panel's three female headers) are bought separately and soldered by the instructor: they carry the field Assembly = hand and are excluded from the JLC BOM and CPL.

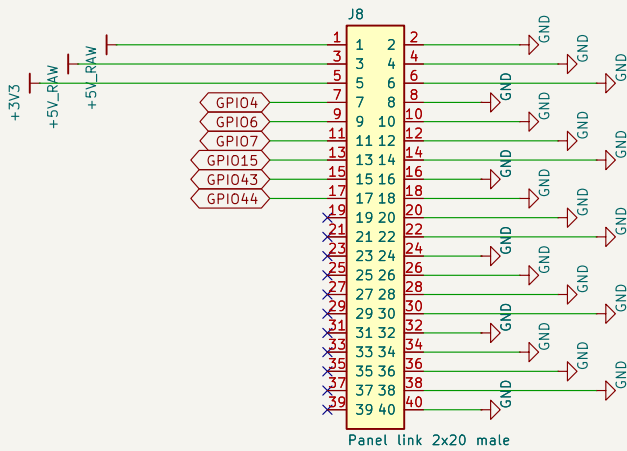
J6 – analog, at x = 12 mm on the bottom side



J7 – digital, at x = 90 mm on the bottom side



J8 – power + spares, at x = 168 mm on the bottom side



J6 pin table

Pins	Signals (odd = signal, even = its return)
1–6	AI1 AGND AI2 AGND AI3 AGND
7–12	AI4 AGND AI5 AGND AI6 AGND
13–18	AI7 AGND AI8 AGND RX AGND
19–24	TX AGND TX AGND AUX AGND
25–30	AO1 AGND AO2 AGND spare AGND
31–36	spare AGND spare AGND spare AGND
37–40	spare AGND spare AGND

J7 pin table

Pins	Signals (odd = signal, even = its return)
1–6	TTL1 GND TTL2 GND TTL3 GND
7–12	TTL4 GND TTL5 GND TTL6 GND
13–18	TTL7 GND TTL8 GND FASTTTL1 GND
19–24	FASTTTL2 GND TRIG_5V GND I2C_SDA GND
25–30	I2C_SCL GND +3V3 GND LED_PWR GND
31–36	LED_WIFI GND LED_ACT GND spare GND
37–40	spare GND spare GND

J8 pin table

Pins	Signals (odd = signal, even = its return)
1–6	+5V_RAW GND +5V_RAW GND +3V3 GND
7–12	GPIO4 GND GPIO6 GND GPIO7 GND
13–18	GPIO15 GND GPIO43 GND GPIO44 GND
19–24	spare GND spare GND spare GND
25–30	spare GND spare GND spare GND
31–36	spare GND spare GND spare GND
37–40	spare GND spare GND

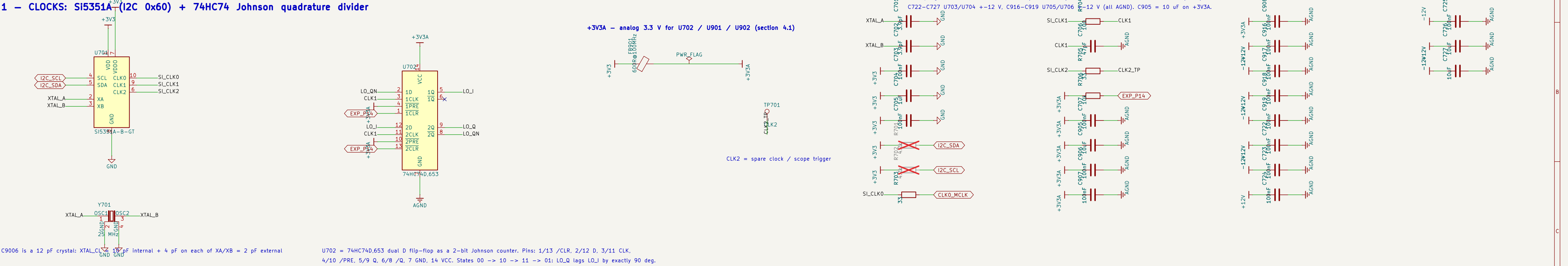
Pin 29 of J7 (LED_PWR) is +3V3 directly; the panel LED series resistor sits on the panel PCB.

Spare pins: J6 has 6 (29–39 odd, AGND partners), J7 has 3 (35–39 odd), J8 has 11 (19–39 odd) – 20 spare signal pins with their returns.

NMR RX (section A) – clocks, tuned tank, LNA, blanking, I/Q commutating mixer, IF

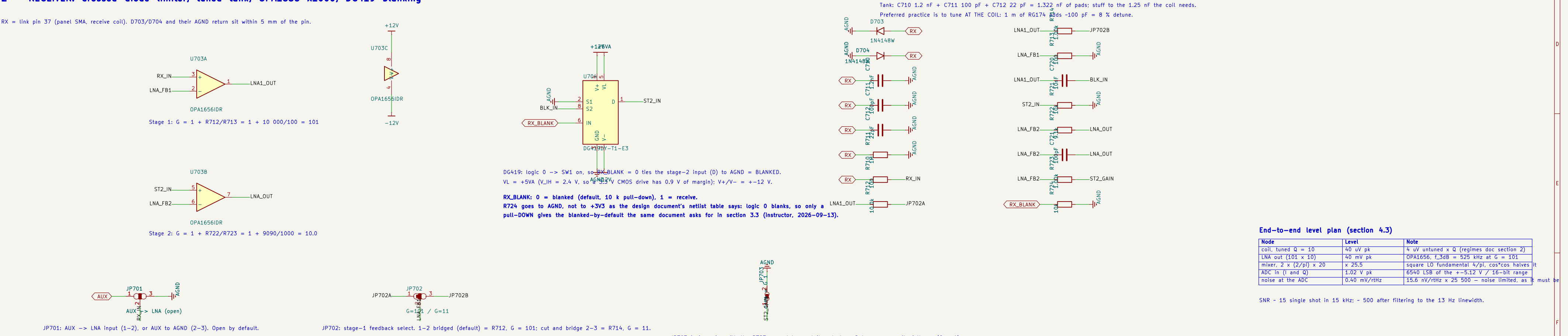
Design coil: 400 turns AWG26, 4 cm x 10 cm, L = 2.53 mH, R_DC = 6.7 R. Tuned with 1.25 nF at f_Larmor = 89.4 kHz (B0 = 2.1 mT): X_L = 1419 R, R_p = Q^2 X_L = 14.2 kOhm at Q = 10, tank noise 15.3 nV/rHz, signal 40 uV pk (v0.7 NMR circuits doc section 0).
LNA: the source impedance is 14.2 kOhm, so CURRENT noise decides. OPA1656 (FET, 6 fA/rHz) gives NF = 0.15 dB; the OPA1612 (1.7 pA/rHz) would cost 5.5 dB = 3.5x more averages. Gain 101 x 10 = 1000 puts 40 mV pk at the mixer and 1.02 V pk on the ADC (section 3.2 / 4.3).
Blanking sits BETWEEN the stages: 60 pC of DG419 charge injection lands on C720 = 10 nF as 6 mV, not on the 1.25 nF tank as 48 mV. Interstage coupling 10 nF / 10 k = 1.59 kHz, so the receiver recovers in 0.3 ms instead of the 300 ms a 1 uF / 100 k would need (section 3.3). RX_BLANK: 0 = blanked (default, 10 k pull-down R724), 1 = receive – an un-driven GPIO leaves the receiver blanked.
Quadrature LO: the Si5351 phase register cannot do 90 deg below 4.72 MHz (PHOFF is 7 bit in units of T_VCO/4, AN619 section 6). U702 is a 2-bit Johnson counter on CLK1 = 4*f_LO; FF1 D = /Q2, FF2 D = Q1 -> LO_L and LO_Q exactly 90 deg apart at any frequency (section 1.4). Pulse /CLR after every PLL change so the counter always starts in state 00.
Mixer supply is +3V3A, NOT +5VA: the TS5A23157 needs V_IH = 0.7*V+ = 3.5 V at 5 V, which a 3.3 V CMOS LO never meets; at 3.3 V it is 2.31 V with 1.0 V of margin (section 4.1). FB901 + 10 uF + 100 nF make +3V3A from +3V3.
Double-balanced commutator: U901 switches S+ / S- into COM (I+ and Q+), U902 the same with the inputs swapped (I- and Q-). The difference amplifiers (G = 20, 0.1 % thin film -> CMRR 54 dB) reject V_MID and the LO-synchronous charge injection; the reference leg returns to AGND so COND_OUT1/2 are centred on 0 V for the ADC's + -5.12 V range (section 4.2).
Grounding (section 8): everything analog on AGND, including U702 (which runs from +3V3A) and the switches, so the commutation instants share the signal ground. The Si5351 is on GND; the single line that crosses is CLK1, damped by R704 / C706. AGND meets GND only at NT1 on the base sheet.

1 – CLOCKS: Si5351A (I2C 0x60) + 74HC74 Johnson quadrature divider

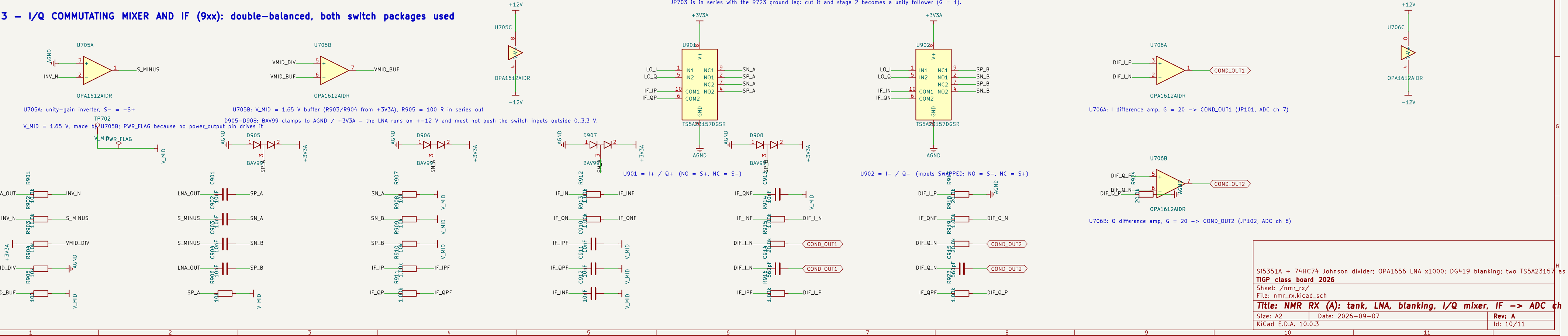


2 – RECEIVER: crossed-diode limiter, tuned tank, OPA1656 x1000, DG419 blanking

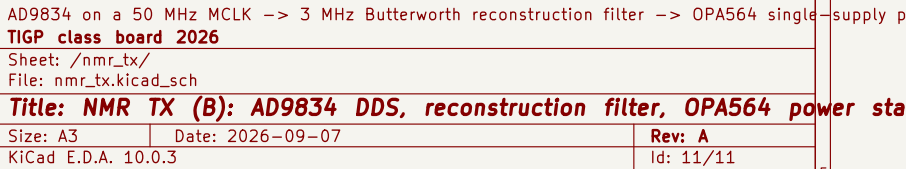
RX = link pin 37 (panel SMA, receive coil), D703/D704 and their AGND return sit within 5 mm of the pin.



3 – I/Q COMMUTATING MIXER AND IF (9xx): double-balanced, both switch packages used



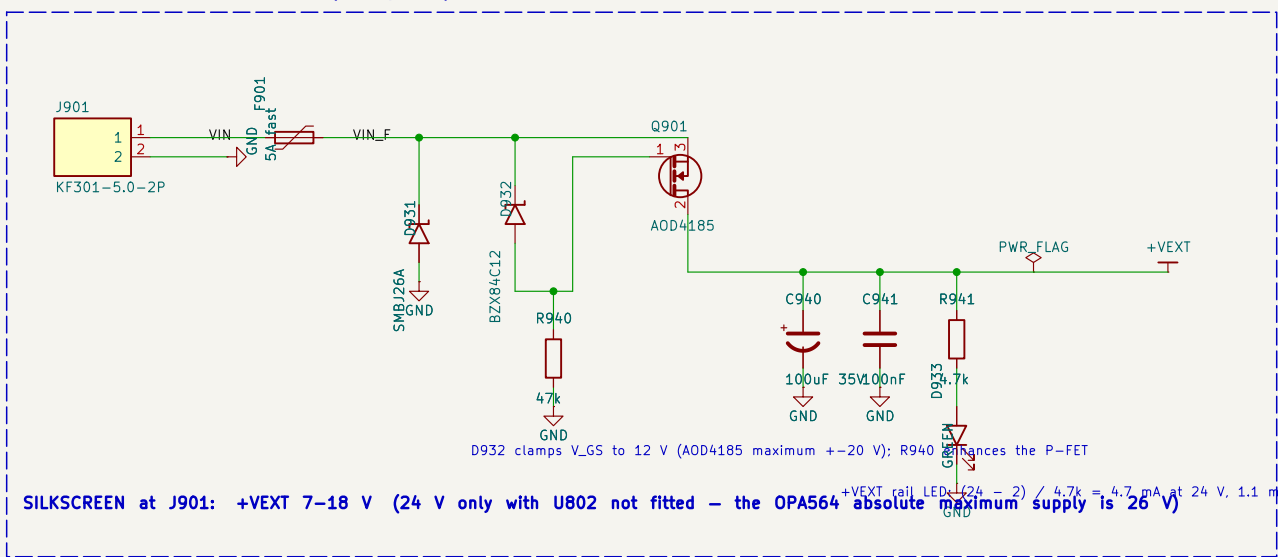
DDS: AD9834BRUZ, MCLK = CLK0_MCLK = 50.000 MHz from the SI5351A on the RX sheet; df = f_MCLK / 2^28 = 0.186 Hz per LSB. R801 6.80 k on FS ADJUST -> I_OUT,FS = 18 x 1.20 V / 6.80 k = 3.18 mA; R802 / R803 200 R loads -> 0.636 V pp at IOUT, inside the 0.8 V output-compliance limit (datasheet pins 19/20). C807 20 pF COG on IOUT prevents clock feedthrough. FSELECT tied to DGND (frequency selected by the FSEL control bit). Reconstruction filter: singly-terminated 3rd-order Butterworth, f_c = 3 MHz, source resistance = the 200 R DAC load, prototype g = 1.5 / 1.3333 / 0.5 -> C808 390 pF, L801 15 uH (LQH32DN150K53L, SRF 26 MHz), C809 130 pF. The nearest image at 48 MHz is down (48/3)^3 = 72 dB; passband -0.05 dB at 2 MHz, flat at 89 kHz. C810 1 uF couples into the power stage (f_HP = 3.2 Hz against the 100 k || 100 k bias divider). Power stage: SINGLE supply +VEXT (7-18 V), V- = GND. E/S, VDIG and both flags are referenced to V-, so V- = GND lets 3.3 V logic drive them directly; on +12 V every one of them would need an optocoupler (SBOS372E Fig. 38). Gain: non-inverting, R807 6.49 k over R808 270 R -> G = 25/50 (15.9 V pp = +-7.95 V = 5.6 mA into the 1419 R coil = t90 417 us at 89.4 kHz). Alternative R808 stuffings: 1.00 k -> G 7.49 and 3.32 k -> G 2.95 for the >= 1 MHz. Protection: R809 11.0 k on ISET -> I_LIM = 1.50 A (NEVER leave ISET open - the datasheet says an open pin damages the device). R810 10 k beats the internal 100 k pull-up on E/S: 3.3 x 10/110 = 0.30 V < 0.8 V = shut down until D801 / D802 S554 clamp the output to +VEXT and to GND; R813 4.7 R 1 W isolates it; R814 / C817 (10 R + 10 nF) is the datasheet snubber; C818 10 uF blocks the 9 V mid-rail out of the coil (Z = 0.18 R at 89 kHz). JP802 + R815 Sequencing (SBOS372E Fig. 36); VDIG (+3V3, from USB) must come up BEFORE V+ (+VEXT, bench supply) - silkscreen note next to the external power input. PowerPAD pin 21 = V- = GND, pad DOWN on the ground pour with a via array (Grounding (re-spec 8.1): the AD9834 AGND and DGND are tied together AT THE DEVICE and belong to the AGND pour; AVDD/DVDD come from +3V3 through FB801 (local rail +3V3D). The OPA564 and the coil return are on GND. C810 is the



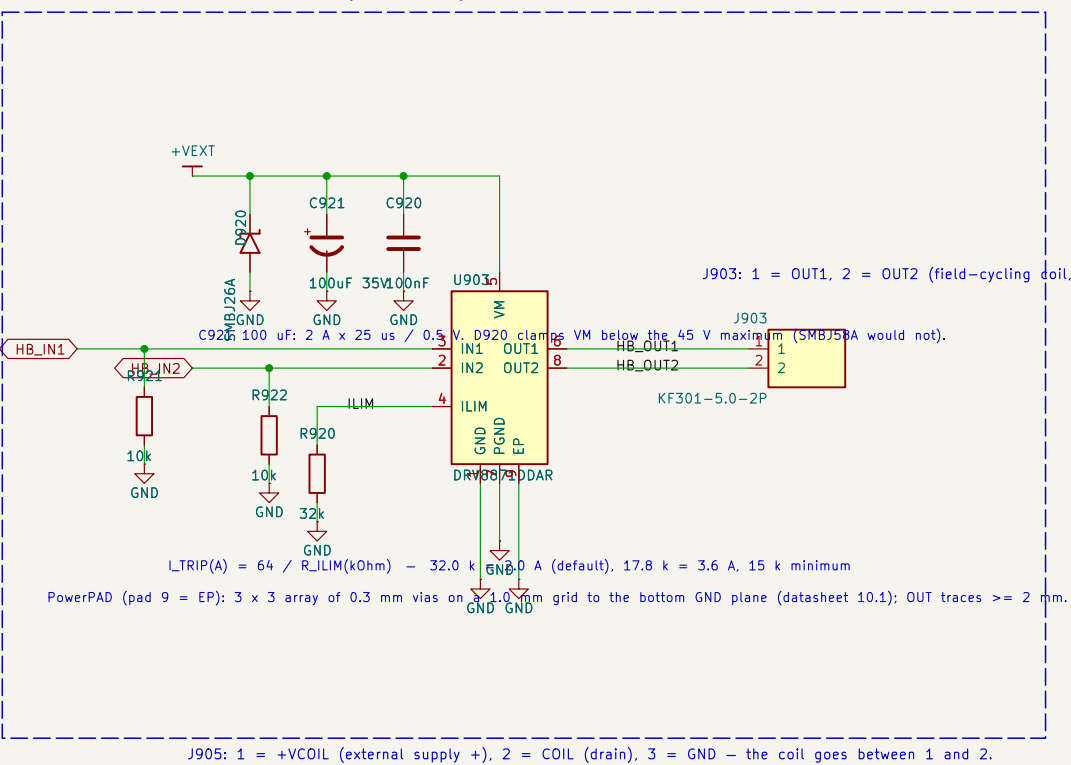
C (9xx) – EXTERNAL POWER INPUT, FIELD-CYCLING H-BRIDGE, POLARIZER SWITCH

Input (7): J901 -> F901 5 A fast (0466005.NRHF) -> D931 SMBJ26A (26 V standoff > the 24 V maximum input; the fuse limits the TVS fault current) -> Q901 AOD4185 reverse-polarity P-FET (15 mOhm: 0.375 W at 5 A against 2.75 W for a Schottky) -> +VEXT. D932 BZX84C12 clamps V_{GS} of Q901 (AOD4185 maximum +−20 V); R940 enhances the P-FET. C940 + C941 are the +VEXT bulk; D933/R941 show the rail. Loads on +VEXT: OPA564 V+ (NMR TX), DRV8871 VM, UCC27517 VDD. Nothing else. H-bridge (5): U903 DRV8871, VM = +VEXT, C921 100 uF is sized for the 25 us internal t_{OFF} window ($C = I \, dt / dV = 2 \, A \times 25 \, us / 0.5 \, V$); C920 100 nF at the pin. R920 sets L_{TRIP} = 64 / R_{ILIM}(kOhm); 1 % because the trip point is directly proportional to it. R921/R922 hold IN1/IN2 low at reset = coast. D920 SMBJ26A sits VM -> GND, NOT across the coil: the bridge has integral body diodes, so the coil freewheels inside the part and the only real risk is supply pumping ($1/2 \, L \, I^2 = 1.8 \, mJ$ into 100 uF from 15 V = 1.2 V rise). The clamp must fire below the DRV8871's 45 V maximum, which is why the brief's SMBJ58A is wrong here. Polarizer (6): FET_GATE -> R930 pull-down + R931 damping -> U904 UCC27517 (4 A driver) -> R932 -> Q904 AOD4184A (40 V, 13 A, 7 mOhm). JP904 picks the driver supply: +VEXT (default, V_{GS} = 12–18 V, R_{DS}(on) at its 10 V spec point) or +5V_{RAW}. Flyback (6.1), one footprint set, three stuffing options: (a) D930 SS54 freewheel, tau = L/R = 2.6 ms – fitted by default; (b) the RC snubber (the Michal route, 2.2 R 5 W + 4700 uF 50 V) over-damped at ~10 ms, +VCOIL <= 24 V – wired EXTERNALLY across J905 pins 1–2, the parts are too big for the board; (c) D934 SMBJ20A fast dump. R933 is the optional 10 mOhm Kelvin shunt in the source return (13.4 A x 10 mOhm = 134 mV = 5 % of the ADS8688 +−2.56 V range, so no divider) brought out as ISENSE_COIL for a spare ADC channel. It is fitted as a 0 R link by default so the source return is never open. Everything on this sheet is on GND (grounding note 8.1/8.2): up to 13 A of polarizer return must never touch the AGND pour. Route the coil pairs tight – TX out and return together, loop area under 1 cm².

EXTERNAL POWER INPUT (design 7): J901 -> F901 -> D931 -> Q901 -> +VEXT



FIELD-CYCLING H-BRIDGE (design 5): DRV8871, VM = +VEXT



SILKSCREEN / WARNINGS (they also go on the board)

USB BEFORE BENCH SUPPLY.
Plug the USB-C cable (or the 5 V jack) in first and let the board boot, then switch on the supply at J901. The gate pull-downs R921/R922/R930 only hold the logic rails up; +VEXT on a dead board leaves the gates floating.

+VEXT 7–18 V
(24 V only with U802 not fitted: the OPA564 maximum supply is 26 V)

+VCOIL <= 24 V with the snubber
+VCOIL <= 12 V with the TVS
(peak drain must stay under the AOD4184A 40 V V_{DS} rating)

All of this sheet is on GND, never AGND. Up to 13 A returns through J905 pin 3.

POLARIZER MOSFET SWITCH (design 6 / 6.1): UCC27517 -> AOD4184A, three flyback options

